



Dysbiosis of Gut Microbiome Is Associated With Rupture of Cerebral Aneurysms

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BACKGROUND AND PURPOSE: Environmental factors are important with respect to the rupture of cerebral aneurysms. However, the relationship between the gut microbiome, an environmental factor, and aneurysm rupture is unclear. Therefore, we compared the gut microbiome in patients with unruptured intracranial aneurysms (UIAs) and ruptured aneurysms (RAs) to identify the specific bacteria causing the rupture of cerebral aneurysms.

METHODS: A multicenter, prospective case-control study was conducted over one year from 2019 to 2020. The fecal samples of patients with stable UIAs and RAs immediately after onset were collected. Their gut microbiomes were analyzed using 16S rRNA sequencing. Subsequently, a phylogenetic tree was constructed, and polymerase chain reaction was performed to identify the specific species.

RESULTS: A total of 28 RAs and 33 UIAs were included in this study. There was no difference in patient characteristics between RAs and UIAs: age, sex, hypertension, dyslipidemia, diabetes status, body mass index, and smoking. No difference was observed in alpha diversity; however, beta diversity was significantly different in the unweighted UniFrac distances. At the phylum level, the relative abundance of *Campylobacter* in the RA group was larger than that in the UIA group. Furthermore, the gut microbiome in the RA and UIA groups exhibited significantly different taxonomies. However, *Campylobacter* was focused on because it is widely known as pathogenic among these bacteria. Then, a phylogenetic tree of operational taxonomic units related to *Campylobacter* was constructed and 4 species were identified. Polymerase chain reaction for these species identified that the abundance of the genus *Campylobacter* and *Campylobacter ureolyticus* was significantly higher in the RA group.

CONCLUSIONS: The gut microbiome profile of patients with stable UIAs and RAs were significantly different. The genus *Campylobacter* and *Campylobacter ureolyticus* may be associated with the rupture of cerebral aneurysms.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: aneurysm ■ gut microbiome ■ inflammation ■ rupture

Unruptured intracranial aneurysms (UIAs) have a prevalence of ≈3%.¹ On rupture, ≈50% of the patients die, and 50% of the surviving patients suffer from severe complications.² Chronic inflammation plays a vital role in cerebral aneurysm (CA) rupture.

Several human CA pathology studies and animal models have demonstrated the presence of inflammatory cells and mediators in the CA wall as well as the subsequent structural deterioration of the CA wall.^{3,4} Chronic inflammation is likely caused by a multifactorial process

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Nonstandard Abbreviations and Acronyms

CA	cerebral aneurysm
OTU	operational taxonomic unit
PPI	proton pump inhibitor
qPCR	quantitative polymerase chain reaction
RA	ruptured aneurysm
SAH	subarachnoid hemorrhage
UIA	unruptured intracranial aneurysm

involving environmental and genetic factors.⁵ While the genetic factors play a moderate role, the environmental factors play a significant role in the cause of subarachnoid hemorrhage (SAH).^{6,7}

Gut microbiota are an environmental factor contributing to human pathophysiology, concerning various diseases where inflammation plays a pivotal role in determination of the disease course and severity. The association between the gut microbiome and cardiovascular diseases such as atherosclerosis has been demonstrated.^{8,9} In CAs, an animal model study demonstrated that the gut microbiome depletion by antibiotics contributed to the pathophysiology of the aneurysm wall through inflammatory modulation.¹⁰ *Hungatella hathewayi* is also associated with the UIA occurrence.¹¹ However, the relationship between the gut microbiome and SAH is unclear.

We hypothesized that there is a difference in the type of microbiome profile between patients with stable UIAs and unstable UIAs that eventually rupture and that this difference may cause chronic inflammation and pathophysiological changes in the aneurysm wall, finally resulting in ruptured aneurysms (RAs). To identify the association between RAs and the microbiome, we compared the gut microbiome in patients with UIAs and RAs immediately after onset and identified the bacteria causing RAs.

METHODS

Standard Protocol Approvals, Registrations, and Patient Consents

Investigators at each institution obtained appropriate approval from the local institutional review board (approval number 846-3), before initiating the case-control study. Each patient was fully informed about the study and provided written consent before participation. For unconscious patients, the next of kin provided written consent. The study is reported in accordance with the STROBE statement.

Study Population

Participants with RAs and UIAs were prospectively recruited between October 2019 and July 2020 from the Osaka

University, Osaka Neurological Institution, Hanwa Memorial Hospital, and Iseikai Hospital.

The inclusion criteria for each group were as follows: intracranial RAs and stable untreated UIAs. We excluded patients aged <18 years, having nonsaccular aneurysmal SAH (dural arteriovenous fistula, cerebral artery dissection, etc), any associated genetic predisposition known to contribute to CA formation (polycystic kidney disease, Ehlers-Danlos syndrome type 4, etc), and antibiotic use within 1 month before admission. The SAH group was divided into 2 groups by grade: those with World Federation of Neurosurgical Societies grade I to III were the low grade group and World Federation of Neurosurgical Societies grade IV to V were the high grade group.

Fecal Sample Collection

For participants with UIAs, fecal samples were collected at home, immediately packed with frozen gel packs within insulated containers and stored at -20°C . By the next day, the sample collection kits were returned by refrigerated shipping keeping at -20°C , and stored at -80°C until processing. This sample collection method was the same as previously described.¹² For participants with RAs, the fecal samples were collected within 48 hours following admission and before the induction of antibiotics to minimize changes in the gut microbial community. Moreover, to prevent rerupture during stool collection, the fecal samples were collected under deep general anesthesia or following curative treatment.

Bacterial DNA Extraction and 16S rRNA Sequencing

DNA from the fecal samples was extracted using an automated DNA extraction machine (GENE PREP STAR PI-480, Kurabo Industries Ltd, Osaka, Japan) following the manufacturer's protocol. The V1–V2 region of the 16S rRNA gene was amplified using the forward primer (16S_27Fmod: TCG GCA GCG TCA GAT GTG TAT AAG AGA CAG AGR GTT TGA TYM TGG CTC AG) and the reverse primer (16S_338R: GTC TCG TGG GCT CGG AGA TGT GTA TAA GAG ACA GTG CTG CCT CCC GTA GGA GT) using the KAPA HiFi Hot Start Ready Mix (Kapa Biosystems, Wilmington, MA).¹³

To sequence the 16S amplicons using the Illumina MiSeq platform (Illumina, San Diego, CA), dual index adapters were attached using the Nextera XT Index kit (Illumina). Each library was diluted to 5 ng/ μL , and equal volumes of the libraries were mixed to a concentration of 4 nmol/L. The DNA of the mixed libraries was quantified by quantitative polymerase chain reaction (qPCR) using the KAPA SYBR FAST qPCR Master Mix (KK4601, Kapa Biosystems) primers, Primer 1 (AAT GAT ACG GCG ACC ACC) and Primer 2 (CAA GCA GAA GAC GGC ATA CGA). The library preparations were performed according to the 16S library preparation protocol by Illumina (Illumina). Libraries were sequenced using the MiSeq Reagent Kit v2 (500 Cycles), 250 bp paired-end.

Microbiome Bioinformatics

The generated FASTQ files were imported, demultiplexed, quality filtered, and processed using the Quantitative Insights into Microbial Ecology 2 pipeline.¹⁴ First, 16S rRNA bacterial partial sequences were qualitatively trimmed and the reads

were truncated into operational taxonomic units (OTUs) referring to Phred quality scores of bases of sequences, using the software package Divisive Amplicon Denoising Algorithm 2 in Quantitative Insights into Microbial Ecology 2 v2020.8, which resulted in a median quality score of 36 (interquartile range, 25–37) for forward reads and 37 (interquartile range, 33–37) for reverse reads. Taxonomy was assigned based on the Silva 138 database release at 99% OTU level. Then, a phylogenetic tree was created to generate the phylogenetic diversity measures using the q2-phylogeny plugin of Quantitative Insights into Microbial Ecology 2, which performed a multiple sequence alignment of the OTU sequences and generated a phylogenetic tree of the OTUs from the alignment.

To even sequence depth, rarefaction was performed to a depth of 26 830 base pairs (90% of the minimal sample depth), which allowed inclusion of all samples.¹⁵ To detect alpha diversity between groups, the output files generated in the previous steps were processed with the *vegan*, *phyloseq*, and *qiime2R* packages in the R v3.3.2 software.¹⁵ Beta diversity was calculated using a phylogenetic tree in weighted and unweighted UniFrac. For assessing the statistical differences in beta diversity metrics between the groups, permutational multivariate ANOVA (PERMANOVA) was performed.

To identify significantly different taxa between the groups, the linear discriminant analysis effect size analysis was performed.¹⁶ The linear discriminant analysis score was used to estimate the effect size of each differentially abundant feature. The threshold of a linear discriminant analysis score was defined as ± 2.0 when comparing the relative taxa abundances between the 2 groups.

Then, a phylogenetic tree of OTUs related to *Campylobacter* was constructed, and identifications at the species level for the OTUs were retrieved from Basic Local Alignment Search Tool (BLAST)-based comparisons against the NCBI 16S reference database. To confirm these results, the extracted genomic DNA was subjected to a specific PCR assay for the genus *Campylobacter* and *Campylobacter* species.

Healthy Control Group

To compare the gut microbiome of healthy people living in the same country, each case with UIA was matched to one control who had neither undergone a surgical procedure nor taken any medicine. The matching was performed from a commercial database (Cykinso Inc, Tokyo, Japan) following the matching factors: residential country, sex, age, and body mass index. Consequently, three UIAs were not matched from the database, and 30 controls were selected. 16S rRNA data were prepared and processed by the aforementioned protocol (Table I in the Supplemental Material). The genus *Campylobacter* presence in the control group was compared with the UIA and RA groups.

Human Microbiome Project Data Set

To compare the gut microbiome of people living in a different residential country, the Human Microbiome Project phase 1 (V1–V3) was used as another control. Processed 16SrRNA data of the Human Microbiome Project phase 1 are available on MicrobiomeDB (<http://microbiomeDB.org>).¹⁷ In the Human Microbiome Project, an age range of 18 to 40 years was employed, and the residential area was urban areas of the United States.¹⁸

The 16S rRNA Gene-Based PCR Assay Specific for *Campylobacter* Species

The extracted genomic DNA was subjected to a PCR assay for detection of *Campylobacter*. Each reaction mixture comprised Q5 Hot Start High-Fidelity 2X Master Mix (New England Biolabs), 500 nmol/L concentrations of the 16S rRNA gene-specific primers (Table II in the Supplemental Material), and 1 ng of genomic DNA in a final volume of 25 μ L. A LifeEco (BIOER, Hangzhou, China) thermal cycler used the following program: 98°C for 5 minutes, followed by 35 cycles of 30 s at 98°C, 30 s at 67°C, and 20 s at 72°C. To visualize the PCR products of the 816-bp region of the 16S rRNA gene, 2 μ L aliquots were electrophoresed using a 2% E-gel EX (Invitrogen).

The Heat Shock Protein (*hsp60*) Gene-Based PCR Assay Specific for *C. ureolyticus*

The *Campylobacter*-positive samples, by the PCR assay, were then subjected to a PCR assay for *C. ureolyticus* identification. Each reaction mixture comprised Q5 Hot Start High-Fidelity 2X Master Mix (New England Biolabs), 500 nmol/L concentrations of the *hsp60* gene-specific primers (Table II in the Supplemental Material), and 1 ng of extracted genomic DNA in a final volume of 25 μ L. The thermal cycler used the following program: 98°C for 5 minutes, followed by 35 cycles of 30 s at 98°C, 30 s at 64°C, and 20 s at 72°C. To visualize the PCR products of the 429-bp region of the *hsp60* gene, 2 μ L aliquots were electrophoresed using a 4% E-gel EX (Invitrogen).

The *hsp60* Gene-Based Real-Time qPCR Assay Specific for *C. concisus* and *C. gracilis*

The *Campylobacter*-positive samples were subjected to a qPCR assay for *C. concisus* and *C. gracilis* identification. Each reaction mixture comprised KAPA SYBR FAST qPCR Master Mix (Kapa Biosystems), 300 nmol/L concentrations of the appropriate primer sets (Table II in the Supplemental Material), and 1 ng of genomic DNA in a final volume of 10 μ L. An Eco 48 real-time qPCR system (PCRmax, Staffordshire, United Kingdom) was used for the qPCR assays with the following program: 95°C for 3 minutes, followed by 40 cycles of 10 s at 95°C, 20 s at 65°C, and 1 s at 72°C. Melting curve analysis was performed under 95°C for 15 s, 55°C for 15 s, and followed by a slow increase from 55°C to 95°C.

Statistical Analysis

The distribution of each continuous variable was assessed by the Shapiro-Wilk W test. Normally distributed parametric data were analyzed using unpaired Student *t* test, and the non-normally distributed nonparametric data were analyzed by the Wilcoxon test. Each categorical variable was analyzed using the χ^2 test or Fisher exact test appropriately. False discovery rate correction using the Benjamin-Hochberg procedure was performed when multiple comparisons were made to evaluate differences between 2 groups. Relative abundance of specific bacteria among multiple groups was analyzed using 1-way ANOVA followed by multiple-comparison Tukey-Kramer test, and the detection rate among multiple groups was analyzed using χ^2 test or Fisher exact test with

Bonferroni correction. Values of $P < 0.05$ or false discovery rate $q < 0.05$ were considered significant. Statistical analyses were performed using JMP software, version 14.0 (SAS Institute, Cary, NC).

Data Availability

Deidentified data will be shared on request from a qualified investigator.

RESULTS

Patients Characteristics

A total of 70 patients were recruited from October 2019 to July 2020. Of these, 8 patients were excluded due to nonaneurysmal SAH and one patient due to inadequate fecal samples. Consequently, 28 RAs and 33 UIAs were included in this study (Figure 1 in the [Supplemental Material](#)). A total of 33 patients with UIAs were followed up for 5.1 ± 3.7 years (interquartile range, 2.4–8.0 years). The characteristics of the patients and aneurysms are summarized in Table 1. The patients' characteristics, including age, sex, hypertension, dyslipidemia, diabetes status, body mass index, and smoking, did not differ in patients with RAs and UIAs. Regarding the aneurysmal characteristics, the distribution of aneurysm location was not different; however, RAs were larger in size than UIAs ($P = 0.0017$).

Table 1. Patients' and Aneurysms' Characteristics

Characteristics	RA, n=28	UIA, n=33	P value
Patient			
Age	65.3 $\pm$ 12.7	69.1 $\pm$ 9.9	0.20
Sex, female	20 (71.4)	25 (75.8)	0.70
HT	16 (57.1)	14 (42.4)	0.25
DL	7 (25.0)	8 (24.2)	0.95
Diabetes	3 (10.7)	6 (18.2)	0.41
CKD	1 (3.6)	0 (0)	0.27
IHD	0 (0)	1 (3.0)	0.35
History of stroke	0 (0)	2 (6.1)	0.19
Smoking	7 (25.0)	5 (15.2)	0.36
Drinking alcohol	7 (25.0)	9 (27.3)	0.84
BMI	21.9 $\pm$ 4.0	22.7 $\pm$ 4.0	0.77
Pre-mRS	0 $\pm$ 0	0.24 $\pm$ 0.83	0.10
Aneurysm			
Size, mm	5.8 $\pm$ 3.2	3.7 $\pm$ 1.6	0.0017*
Location			0.36
Anterior circulation	27 (96.4)	29 (87.9)	
Posterior circulation	1 (3.6)	4 (12.1)	
Phases score	8.43 $\pm$ 1.91	7.06 $\pm$ 1.14	0.0019*

Continuous variables are shown as mean $\pm$ SD and categorical variables are shown as n (%). BMI indicates body mass index; CKD, chronic kidney disease; DL, dyslipidemia; HT, hypertension; IHD, ischemic heart disease; mRS, modified Rankin Scale; RA, ruptured aneurysm; and UIA, unruptured intracranial aneurysm.

*Statistical significance.

Differences in Gut Microbiome Between the RA and UIA Groups

The OTU data set for RA and UIA group comprised 6529 OTUs that classified 314 genera, 104 families, 56 orders, 25 classes, and 12 phyla. Between the RA and UIA groups, no difference was observed for alpha diversity, which includes species richness (represented by Chao1, observed species, and abundance-based coverage estimator) and richness and evenness (represented by Shannon index, Simpson index, and inverse Simpson index) of the microbial community (Figure 1A). In beta diversity, a significant difference was observed in the unweighted UniFrac distances ($r^2 = 0.023$, $P = 0.020$, Figure 1B).

The dominant phyla of all patients were the *Firmicutes* and *Bacteroides*, $52.2 \pm 11.5\%$ and $36.5 \pm 13.8\%$, respectively (Table 2 and Figure 1C). At the phylum level, the relative abundance of *Campylobacteriota* was significantly higher in the RA group (false discovery rate, $q = 0.007$; Table 2).

In the linear discriminant analysis effect size analysis, we identified 3 bacterial phyla, 2 classes, 3 orders, 9 families, and 28 genera that were significantly different in their relative abundance in the RA and UIA groups (Figure 2). For the analysis of the abundance ratio of these 28 genera, 23 genera were enriched in the RA group. The relative abundance of these genera was not different among the different hospitals (Figure II in the [Supplemental Material](#)). Of the 23 genera, *Campylobacter* is commonly identified as a genus of pathogenic bacteria and was focused upon in this study.

Identification of the Genus *Campylobacter* and *Campylobacter* Species by PCR

The genus *Campylobacter* were classified into species, *C. ureolyticus*, *C. concisus*, *C. gracilis*, and *C. hominis* (Figure 3C). To elucidate these results of 16S rRNA, PCR for specific bacteria was performed. The genus *Campylobacter* and *Campylobacter* species, identified by the 16S rRNA, were also PCR-positive. In some patients, the genus *Campylobacter* and *Campylobacter* species that were not identified by the 16S rRNA were also found positive by the PCR assay. As a result, the number of positive samples of the genus *Campylobacter* and *C. ureolyticus* in the RA group was significantly larger than that of the UIA group by PCR and 16S rRNA (Table 3). The number of positive samples of the genus *Campylobacter* was larger in women than men by PCR, unlike 16S rRNA (85.0% versus 37.5%, $P = 0.007$, Figure III in the [Supplemental Material](#)).

Difference of Gut Microbiome in Severity

Of the 28 RAs, 16 were low-grade and 12 were high-grade. No differences in the patients' and aneurysms' characteristics were observed in patients with high-grade SAH and low-grade SAH (Table III in the [Supplemental Material](#)).

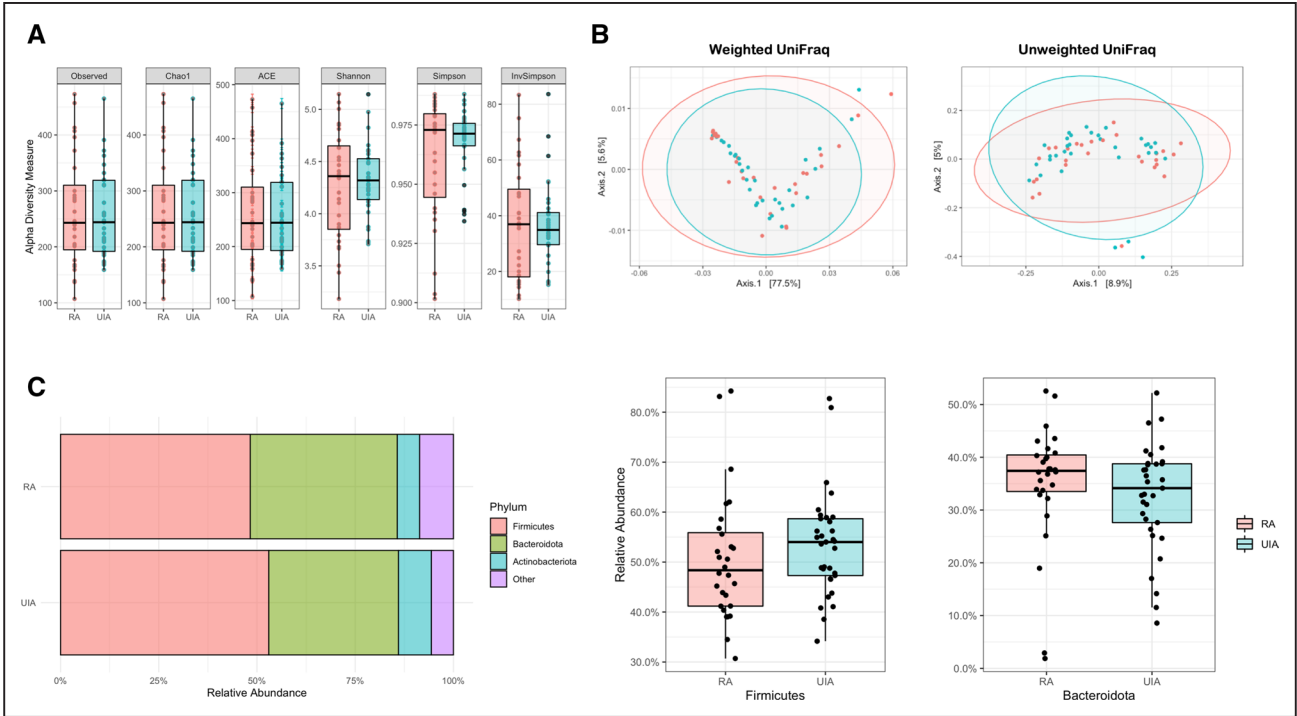


Figure 1. Comparison of microbial diversity and microbiome composition between ruptured aneurysm (RA) and unruptured intracranial aneurysm (UIA) groups.
A, No significant difference was found in alpha diversity based on the observed, Chao1, abundance-based coverage estimator, Shannon, Simpson, and inverse Simpson indices ($P>0.05$). **B**, Principal coordinate analysis illustrating the grouping patterns of the RA and UIA groups based on the weighted and unweighted UniFrac distances. There was no difference in beta diversity with weighted UniFrac distances ($R^2=0.0046$, $P=0.81$), but was a significant difference in beta diversity with unweighted UniFrac distances ($R^2=0.023$, $P=0.013$). **C**, Distribution of the relative abundance of bacteria at the phylum level. The relative abundance of *Firmicutes* and *Bacteroides* did not differ significantly between the groups.

Furthermore, there was no difference in the aforementioned 3 bacterial phyla, 2 class, 3 order, 9 family, and 30 genera between the low and high grade (Figure IV in the Supplemental Material).

DISCUSSION

In the present study, we investigated the gut microbiome profiles of patients with stable UIAs and RAs and

the specific bacteria that cause RAs using 16S rRNA sequencing and PCR assay to identify the association between the gut microbiome and the rupture of CAs. At the phylum level, the relative abundance of *Campylobacter* in the RA group was significantly larger than that in the UIA group. In the linear discriminant analysis effect size analysis, several taxonomic profiles of the gut microbiome in the UIAs and RAs groups were significantly different. Particularly, *Campylobacter* is generally known as

Table 2. Relative Abundance at the Phylum Level

Relative abundance	RA	UIA	P value	q value
Firmicutes, %	50.6±2.4	53.5±1.8	0.16	0.32
Bacteroidetes, %	35.1±2.2	32.6±1.8	0.16	0.32
Actinobacteria, %	6.6±0.93	8.4±1.5	0.77	0.96
Proteobacteria, %	6.2±0.84	4.4±0.79	0.072	0.18
Fusobacteriota, %	0.68±0.27	0.76±0.41	0.87	0.87
Desulfobacteria, %	0.48±0.093	0.21±0.047	0.044*	0.15
Synergistota, %	0.17±0.11	0.0060±0.00030	0.46	0.66
Campylobacteriota, %	0.067±0.027	0.0054±0.00041	0.0007*	0.007*
Verrucomicrobiota, %	0.062±0.034	0.11±0.090	0.86	0.96
Cyanobacteria, %	0.061±0.057	0±0	0.026*	0.13

Continuous variables are shown as mean±SEM. q values were derived on the basis of the false discovery rate. RA indicates ruptured aneurysm; and UIA, unruptured intracranial aneurysm.
*Statistical significance.

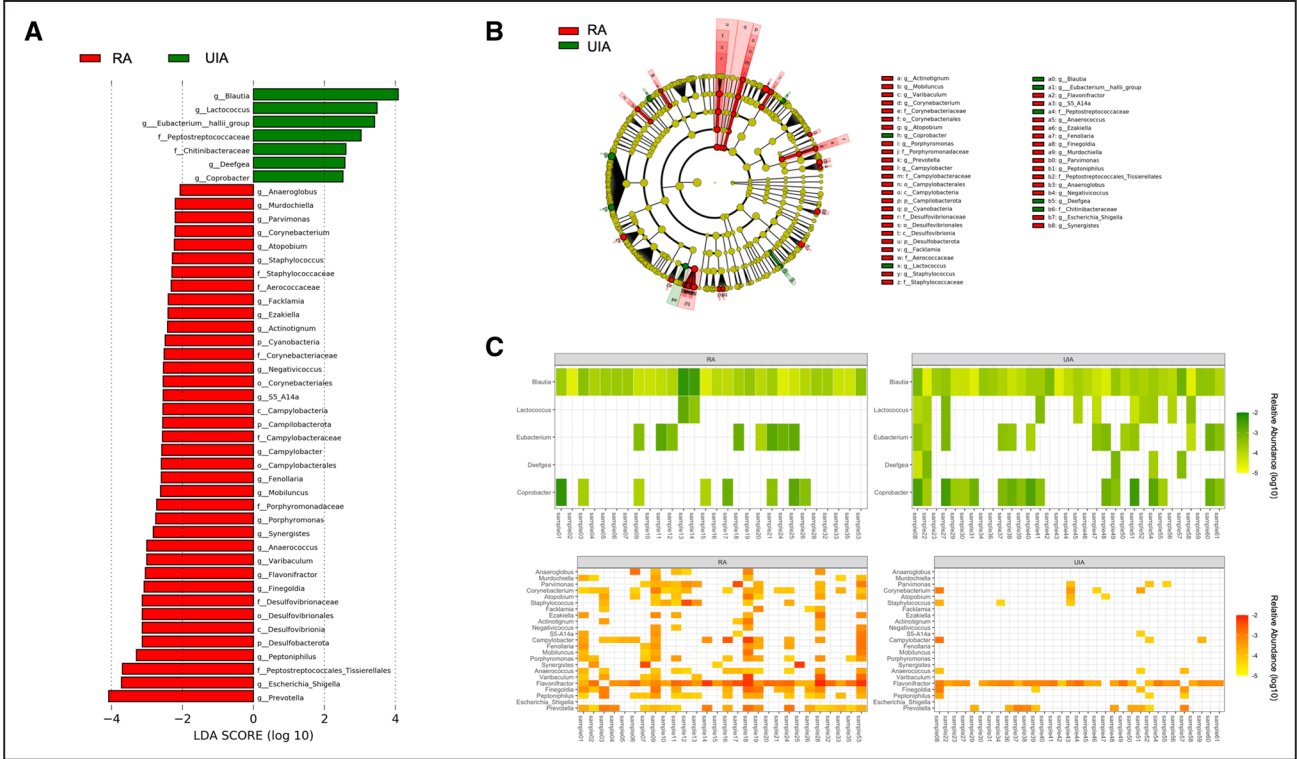


Figure 2. Discriminative taxa between the ruptured aneurysm (RA) and unruptured intracranial aneurysm (UIA) groups. **A**, Discriminative taxa between RA and UIA group were determined using linear discriminant analysis effect size (LEfSe). The LEfSe analysis revealed that 3 bacterial phyla, 2 class, 3 order, 9 family, and 28 genera were significantly different between RA and UIA group. The green bar chart represents the bacteria that were more abundant in the UIA group and the red bar chart represents the bacteria that were more abundant in the RA group. **B**, The cladograms report the taxa showing different abundance values according to LEfSe. In the RA group, *Campylobacter* was more abundant from the phylum level to the genus level. **C**, Heat-map of the discriminative taxa revealed by LEfSe. Most of the discriminative taxa in the RA group did not exist in the UIA group.

infectious disease-causing among these bacteria.^{19,20} Compared with the control groups of the present study (same country; Figure V in the Supplemental Material) and another control of the Human Microbiome Project data set (different country), the UIA group had a similarly low detection ratio of the genus *Campylobacter* and the

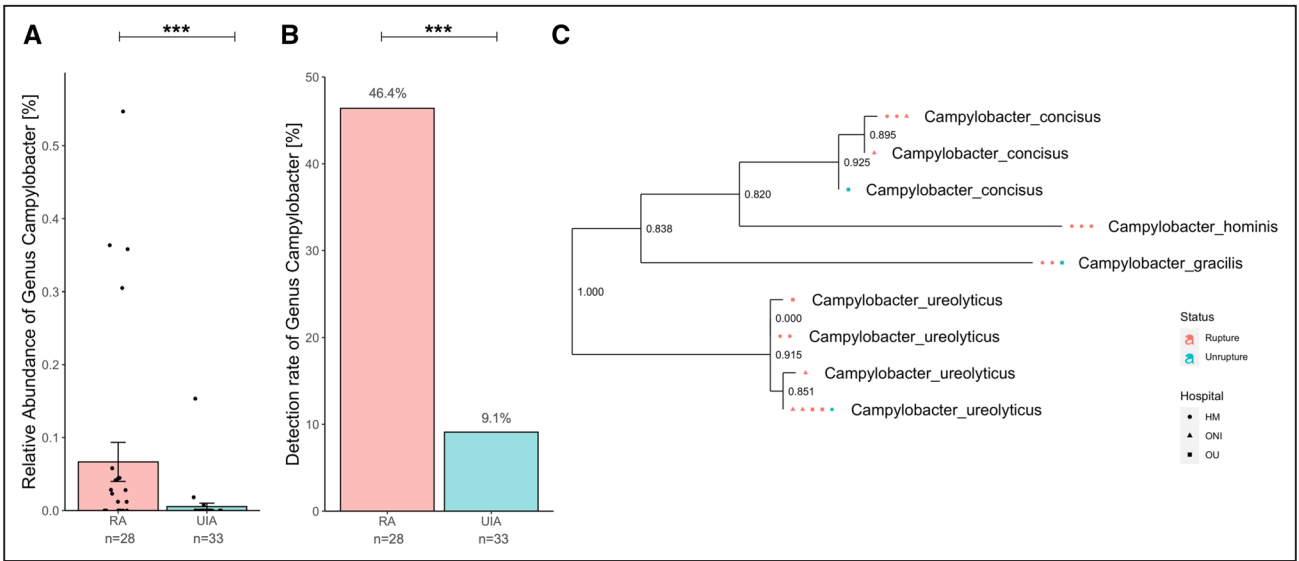


Figure 3. *Campylobacter* between the ruptured aneurysm (RA) and unruptured intracranial aneurysm (UIA). **A**, Relative abundance of the genus *Campylobacter* was significantly larger in the RA groups than the UIA group ($P=0.0007$). **B**, Detection rate of the genus *Campylobacter* in the RA group was higher than in the UIA groups ($P=0.00013$). *** $P<0.001$. **C**, Phylogenetic tree based on *Campylobacter* by the 16S rRNA. The genus *Campylobacter* was classified into *C. concisus*, *C. hominis*, *C. gracilis*, and *C. ureolyticus* and identified in 3 hospitals.

Table 3. The Number of Positive Samples by the 16S rRNA and PCR

Species	16S rRNA				PCR			
	RA, n=28	UIA, n=33	P value	q value	RA, n=28	UIA, n=33	P value	q value
Genus <i>Campylobacter</i>	13 (46.4)	3 (9.1)	0.0010*	0.0050*	20 (71.4)	9 (27.3)	0.00058*	0.0012*
<i>C. ureolyticus</i>	7 (25.0)	1 (3.0)	0.011*	0.028*	11 (39.3)	1 (3.0)	0.00039*	0.0016*
<i>C. concisus</i>	4 (14.3)	1 (3.0)	0.17	0.21	4 (14.3)	3 (9.1)	0.69	0.69
<i>C. gracilis</i>	2 (7.1)	1 (3.0)	0.59	0.59	3 (10.7)	2 (6.1)	0.51	0.68
<i>C. hominis</i>	3 (10.7)	0 (0.0)	0.091	0.15	...†	...†	...†	...†

Categorical variables are shown as n (%). *q* value were derived on the basis of the false discovery rate. PCR indicates polymerase chain reaction; RA, ruptured aneurysm; and UIA, unruptured intracranial aneurysm.

*Statistical significance.

†Methods for detection *C. hominis* has not been established and PCR was not performed.

RA group had a significantly higher ratio (Figure VI in the [Supplemental Material](#)). Thus, the presence of *Campylobacter* spp. is usually rare, and *Campylobacter* spp. may be associated with RAs. Then, a phylogenetic tree of OTUs related to *Campylobacter* was constructed, and 4 species were identified. To clarify these results, we identified the *Campylobacter* spp. using PCR. We found that the genus *Campylobacter* and *C. ureolyticus* were significantly increased in the RA group as well as the 16S rRNA.

Campylobacter Is Specific to RAs

To date, there are no previous reports on SAH and the gut microbiota in humans. In animal models, studies suggest that intestinal bacteria may be involved in the rupture of CA.^{10,11} These findings suggest that the gut microbiome is likely involved in the rupture of CA. Ideally, we should compare the intestinal microbiota before rupture with that of UIAs to identify the bacteria responsible for rupture, but it is difficult given the low rupture rate. Therefore, we compared the gut microbiome immediately following the rupture with that of long-term stable UIAs in this study. However, brain damage itself, such as ischemic stroke and even mild traumatic brain injury, is already widely known to affect the gut microbiome,^{21,22} which would be true in SAH. This is because the communication between gut and brain is regulated by mediators such as immune modulatory metabolites, gut peptides, neurotransmitters, and cytokines.²³ Therefore, it is unclear whether the difference between the groups was affected by the stress of a SAH or had already occurred before the onset of the SAH. A study comparing patients with stroke with normal patients found that 7 genera, including unclassified *Enterobacteriaceae*, were more abundant in the stroke group, and 11 genera, including *Prevotella* and *Lachnospira*, were more abundant in the healthy group.^{22,24} Also in this study, the *Enterobacteriaceae* family (*E. shigella*) was more abundant in the SAH group, and the *Lachnospiraceae* family (*blautia*) was more abundant in the healthy group. Therefore, most bacteria detected in this study were considered to be affected by the SAH-induced stress. However, *Campylobacter* levels did not increase in patients with ischemic stroke.^{22,24} In a previous study

comparing systemic inflammatory response syndrome with control patients at our hospital, the *Campylobacter* did not increase.²⁵ We suggest that the genus *Campylobacter*, rather than SAH-induced stress, may be specific to RAs.

Risk Factor for Campylobacter Infection

C. ureolyticus is the second most common causative agent of *Campylobacter*-related gastroenteritis. Furthermore, *Campylobacter* is implicated in ulcerative colitis, bacterial vaginosis, and lesions of the female genital tract.²⁶ *C. ureolyticus* is more common in older women.²⁶ In several studies, PPI (proton pump inhibitor) use has been reported as a risk factor for *Campylobacter* infection.²⁷ Furthermore, many drugs including PPI are known to change gut microbiome composition.²⁸ However, in the present study, there was no difference in drugs between RA and UIA group and no relationship between drugs and *Campylobacter* (Table IV in the [Supplemental Material](#)). In a recent study, alcohol consumption frequency and bowel movement quality were reported to strongly affect the distribution of gut microbiome; the study showed increased robustness and reproducibility in resolving the members of the gut microbiota that are truly associated with human disease by even matching of these variables between cases and controls.²⁹ In the present study, there was no difference in drinking alcohol and taking laxatives, and while information on bowel movement quality could not be collected in detail, this study was considered to be highly reproducible.

Mechanism Underlying the Rupture of CAs due to Campylobacter Infection

In an in vitro study, cells infected with *C. ureolyticus* produced significantly higher levels of interleukin-8 than the negative controls.³⁰ In a human study, MMP-8 (matrix metalloproteinase 8), MMP-9, myeloperoxidase, human neutrophil elastase, and tissue inhibitor of MMP-1 were 2-3 times higher in *Campylobacter*-infected patients compared with healthy controls.³¹ Thus, *Campylobacter* can induce an inflammatory response. On the contrary,

several studies of CAs have found that inflammation, neutrophils, and cytokines had a crucial role in CA progression and rupture.^{3,32} Thus, systemically enhanced cytokines, neutrophil-derived proteolytic and oxidative stress by *Campylobacter* may promote vascular remodeling and cell death of CA walls, and eventually, combined with hemodynamics and genetics, lead to changing UIA into RA.

Research on CAs remains challenging. The rupture rate of aneurysms is higher in Japanese people; however, the underlying association is still unclear.⁵ There is currently no evidence of higher rates of rupture in East Asians with a similar genetic lineage to the Japanese. If a genetic factor is not implicated in the rupture of CAs, environmental factors may be more likely. In several studies, a greater diversity of the gut microbiome has been reported among populations.³³ Another study reported that the gut microbiome of the Japanese population is considerably different from that of other populations.³⁴ In the present study, we demonstrated that *Campylobacter* spp. and *C. ureolyticus* may be associated with RAs. Thus, the identification of gut bacteria as a novel environmental factor for the rupture of CAs is noteworthy.

This study had several limitations. First, other risk factors, including genetics and metabolomics, were not investigated. Further studies, including animal models and longitudinal human studies, are needed to determine the cause-effect relationship between the gut microbiome and rupture of the CA. Second, this study was conducted in a single country, and it is important to note the regional and racial differences when assimilating the current results, in comparison with other studies. Third, *Campylobacter* is widely known to be usually acquired by diet, however, dietary information was not collected. Finally, information about the time after rupture was not collected precisely, and it was not clear how the time until sampling affected the detection rate and abundance of *Campylobacter*. In the present study, *Bacteroides-Firmicutes* ratio was not different between the RA and UIA group, which indicated that we could collect samples appropriately without dynamic change.³⁵

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Disclosures

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Supplemental Materials

Supplemental Figures I–VI
Supplemental Tables I–IV

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